

Design and Implementation of a Smart Courier and Parcel Management System

Presented by

Mst. Sharmin Khatun

2223081010

Tasnim Jahan Mim

2221081138

Department of CSE
UTTARA UNIVERSITY

Supervised by

Pranta Banik

Lecturer

Department of CSE
UTTARA UNIVERSITY

Outline

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Objective

- ✓ Develop a secure role-based courier management system using the MERN stack.
- ✓ Enable real-time parcel tracking and status updates using Socket.IO and maps.
- ✓ Provide admin analytics and reporting tools for efficient delivery management.

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Existing Problem

- ❑ Courier services rely on manual or fragmented systems for parcel management.
- ❑ Customers do not get real-time visibility of parcel delivery status.
- ❑ Delivery agent coordination and assignment are inefficient.
- ❑ Lack of centralized monitoring and performance analytics for administrators.

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Motivation

- ❖ Many courier services still rely on manual and fragmented systems.
- ❖ Customers lack real-time visibility of parcel delivery status.
- ❖ Delivery coordination becomes inefficient without centralized control.
- ❖ A web-based system can improve transparency and operational efficiency.

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Related Works

- ❑ [1] **Ahmed et al. (2019)** proposed a web-based courier management system, but parcel tracking is not fully automated and updates are delayed.
- ❑ [2] **Rahman et al. (2020)** developed an online delivery platform; however, customers cannot monitor real-time parcel status.
- ❑ [3] **Khan et al. (2021)** focused on logistics operations, yet efficient route optimization for delivery agents was not implemented.

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Gap Analysis

Challenges & Solution:

■ Challenge 1: Manual Tracking and Delays.

- ❖ Many parcels are tracked manually, causing errors and delayed updates.
- ❖ Solution: Implement an automated tracking system with barcode scanning or QR codes to update parcel status in real-time.

■ Challenge 2: Lack of Customer Visibility

- ❖ Customers cannot see the real-time status of their parcels, leading to complaints and calls.
- ❖ Solution: Provide customer notifications via SMS, email, or mobile app showing parcel status and estimated delivery time.

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Methodology

❑ **System Design & Architecture:**

- Design a role-based MERN architecture separating frontend, backend, and database layers.

❑ **Backend Development**

- Implement REST APIs, JWT authentication, role-based access control, and real-time communication using Socket.IO.

❑ **Frontend Development**

- Build responsive user interfaces for Admin, Agent, and Customer using React.

❑ **Real-Time Tracking & Mapping**

- Integrate live status and location updates with Socket.IO and visualize them using OpenStreetMap and Leaflet.

❑ **Testing & Deployment**

- Test system functionality and deploy the application for academic demonstration.

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Expected Outcomes

- ❑ Faster and Accurate Parcel Handling – Automation reduces manual errors and speeds up processing.
- ❑ Improved Customer Experience – Real-time tracking and notifications keep customers informed and satisfied.
- ❑ Better Management Insights – Centralized data enables reporting on deliveries, delays, and operational performance.

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Timeline

Phase	Tasks	Duration
Planning	Literature Review & Gap Analysis	Weeks 1-2
Design	Database Schema & UI Wireframing	Weeks 3-4
Development	Backend API & Frontend Integration	Weeks 5-7
Testing	Bug Fixing & User Acceptance Testing	Weeks 7-9
Deployment	Final Hosting & Documentation	Week 9-12

Conclusion

- ✓ Automation Improves Efficiency – The system reduces manual work and errors in parcel handling.
- ✓ Customer Satisfaction Increases – Real-time tracking and notifications enhance transparency and trust.
- ✓ Data-Driven Decisions Are Possible – Centralized data and reporting help management monitor and optimize operations.

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